👩🏾‍🏫 Staff Guide · Small Engines Overhaul · Unit 6 Ignition (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. No slide decks: students follow the class teacher's lesson and slides with Level 3 readings, guided notes and a quick check.

1. How the pieces line up

Gen ed lesson	Level 3 piece
His readings and worksheets	Level 3 reading (5 short paragraphs) + check questions
Class notes	Level 3 guided notes + key
Unit quiz / test	Level 3 quick check + key
Review (Kahoot / Jeopardy)	1 Level 3 game on the class site: 10 questions + a matching round

2. Supports built in

- 🔊 Read aloud on the reading and notes pages (tap the button)
- Key terms listed at the top of the notes
- Short-answer lines sized for 1-2 sentences

3. What to grade

- Notes and reading checks: same points as his worksheets for that lesson
- Quick check: score, then scale to his test points
- Modified grading only as the IEP allows. Check with the case manager.

4. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **Ignition Handout**: Title only in Canvas; our notes and quiz are team-written.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

⚡ Ignition: Guided Notes

Small Engines Overhaul · Unit 6 · Level 3 (closer to the class work) · goes with the Ignition Handout and magneto ignition topic

Name: _____

Use the class slides and your reading. Key words:

magneto *armature* *primary winding* *secondary winding* *spark plug gap* *air gap* *flywheel key* *kill switch* *spark tester*

A. Magneto

1. Explain how the magneto makes high voltage.

_____

_____

_____

B. Gaps

2. The plug gap is checked with a _____.

3. The armature air gap is set with a _____.

C. Diagnosis

4. Fill in the table.

Plug condition	Likely cause
Black, sooty	
White, blistered	
Light tan	

⚡ Ignition: Reading

Small Engines Overhaul · Unit 6 · Level 3 (closer to the class work) · goes with the Ignition Handout and magneto ignition topic

Name: _____

Ignition: Making the Spark

The power stroke begins with a spark. In most small engines, the spark comes from a magneto, a system that makes its own electricity, so no battery is needed. Permanent magnets are cast into the flywheel. Each time they pass the armature, a coil mounted beside the flywheel, they create a current.

The armature has two windings. The primary winding makes a small current; the secondary winding, with many more turns of wire, boosts it to thousands of volts. This high voltage travels through the plug wire to the spark plug and jumps the gap between the center and side electrodes, igniting the compressed mixture.

Two gaps must be set to the manual's spec: the spark plug gap, checked with a gap gauge, and the armature air gap between the coil and flywheel, set with a gauge card. The flywheel key, a soft metal key, positions the flywheel for correct timing. If a mower blade hits a rock, the key can shear, causing hard starting or kickback.

The kill switch stops the engine by grounding the coil. To check the ignition, disconnect the plug wire and use a spark tester; a bright blue spark is good. Reading the spark plug's color also helps diagnose the engine.

Check your reading

1. Why doesn't a small engine need a battery for spark?

2. What two gaps must be set?

3. What does a sheared flywheel key cause?

⚡ Spark Plug & Spark Check: Lab Sheet

Small Engines Overhaul · Unit 6 · Level 3 · goes with Ignition Handout

Name: _____

Question: Is the engine's ignition system making a good spark, and what does the spark plug show?

Materials

plug socket, gap gauge, spark tester, gauge card, plug chart

⚠️ Safety

- Safety glasses
- Plug wire disconnected
- Use the spark tester; never hold the wire
- Tools with adult supervision

Prediction

Predict the plug's color and what it means.

_____

_____

Procedure

1. Remove plug; read color.
2. Measure and set gap.
3. Reinstall plug.
4. Spark test.
5. Check air gap.

Data

Check	Result	Spec / normal
Plug color		light tan
Plug gap		
Spark color		bright blue
Air gap		

Analysis (Claim, Evidence, Reasoning)

1. Claim: is the ignition OK? Evidence: spark and gaps. Reasoning: what the plug color says about the fuel mixture.

⚡ Ignition: Quick Check

Small Engines Overhaul · Unit 6 · Level 3 (closer to the class work) · goes with the Ignition Handout and magneto ignition topic

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Where are the magneto magnets?

A. In the spark plug | B. In the carburetor | C. In the flywheel

2. The armature air gap is set with…

A. A gauge card | B. A hammer | C. A micrometer

3. A sheared flywheel key affects…

A. Fuel level | B. Oil pressure | C. Ignition timing

4. The kill switch works by…

A. Grounding the coil | B. Closing the choke | C. Draining fuel

5. Describe how to safely test for spark.

6. What does a black, sooty spark plug tell you?

⚡ Ignition: Guided Notes · ANSWER KEY

Small Engines Overhaul · Unit 6 · Level 3 (closer to the class work) · goes with the Ignition Handout and magneto ignition topic

Name: _____

Use the class slides and your reading. Key words:

magneto · *armature* · *primary winding* · *secondary winding* · *spark plug gap* · *air gap* · *flywheel key* · *kill switch* · *spark tester*

A. Magneto

1. Explain how the magneto makes high voltage.
Flywheel magnets pass the armature; the primary makes current and the secondary steps it up to thousands of volts.

B. Gaps

2. The plug gap is checked with a **gap gauge** .

3. The armature air gap is set with a **gauge card** .

C. Diagnosis

4. Fill in the table.

Plug condition	Likely cause
Black, sooty	**rich mixture**
White, blistered	**lean/overheating**
Light tan	**normal**

🔑 *Key written by our team from the class teacher's worksheets and slides, please check.*

⚡ Spark Plug & Spark Check: Lab Sheet · ANSWER KEY

Small Engines Overhaul · Unit 6 · Level 3 · goes with Ignition Handout

Name: _____

Question: Is the engine's ignition system making a good spark, and what does the spark plug show?

Materials

plug socket, gap gauge, spark tester, gauge card, plug chart

⚠️ Safety

- Safety glasses
- Plug wire disconnected
- Use the spark tester; never hold the wire
- Tools with adult supervision

Prediction

Predict the plug's color and what it means.

Answers vary.

Procedure

1. Remove plug; read color.
2. Measure and set gap.
3. Reinstall plug.
4. Spark test.
5. Check air gap.

Data

Check	Result	Spec / normal
Plug color	**black**	light tan
Plug gap	**0.028**	**0.030**
Spark color	**blue**	bright blue
Air gap	**0.010**	**0.010**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: is the ignition OK? Evidence: spark and gaps. Reasoning: what the plug color says about the fuel mixture.
Spark is good and gaps in spec; black plug means rich, so check choke/air filter.

🔑 *Key written by our team from the class teacher's lab sheets, please check.*

⚡ Ignition: Quick Check · ANSWER KEY

Small Engines Overhaul · Unit 6 · Level 3 (closer to the class work) · goes with the Ignition Handout and magneto ignition topic

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Where are the magneto magnets?

A. In the spark plug | B. In the carburetor | **C. In the flywheel**

2. The armature air gap is set with…

A. A gauge card | B. A hammer | C. A micrometer

3. A sheared flywheel key affects…

A. Fuel level | B. Oil pressure | **C. Ignition timing**

4. The kill switch works by…

A. Grounding the coil | B. Closing the choke | C. Draining fuel

5. Describe how to safely test for spark.

Disconnect plug wire, attach a spark tester, pull the rope, look for bright blue spark.

6. What does a black, sooty spark plug tell you?

The mixture is too rich (too much fuel), e.g., choke or dirty air filter.

🔑 *Written by our team from his unit calendar, worksheets and review games, please check.*